

Effect of endurance exercise training and curcumin intake on central arterial hemodynamics in postmenopausal women: pilot study.



BACKGROUND: Lifestyle modification (i.e., regular physical activity and diet) is effective in preventing the age-related increase in cardiovascular disease risks. Potential therapeutic effects of curcumin (diferuloylmethane) have been confirmed on various diseases, including cancer and Alzheimer's disease, but the effects of curcumin have not been tested on central arterial hemodynamics. The aim of this pilot study was to test the hypothesis that the regular endurance exercise combined with daily curcumin ingestion lowers the age-related increase in left ventricular (LV) afterload to a greater extent than monotherapy with either intervention alone in postmenopausal women using a randomized, double-blind, placebo-controlled, parallel manner.

METHODS: Forty-five women were randomly assigned to four interventions: "placebo ingestion" (n = 11), "curcumin ingestion" (n = 11), "exercise training with placebo ingestion" (n = 11), or "exercise training with curcumin ingestion" (n = 12). Curcumin or placebo pills (150 mg/day) were administered for 8 weeks. Aortic blood pressure (BP) and augmentation index (AIx), an index of LV afterload, were evaluated by pulse wave analysis from tonometrically measured radial arterial pressure waveforms.

RESULTS: There were no significant differences in baseline hemodynamic variables among four groups. After the interventions, brachial systolic BP (SBP) significantly decreased in both exercise-trained groups ($P < 0.05$ for both), whereas aortic SBP significantly decreased only in the combined-treatment (e.g., exercise and curcumin) group ($P < 0.05$). Heart rate (HR) corrected aortic AIx significantly decreases only in the combined-treatment group.

CONCLUSIONS: These findings suggest that regular endurance exercise combined with daily curcumin ingestion may reduce LV afterload to a greater extent than monotherapy with either intervention alone in postmenopausal women.